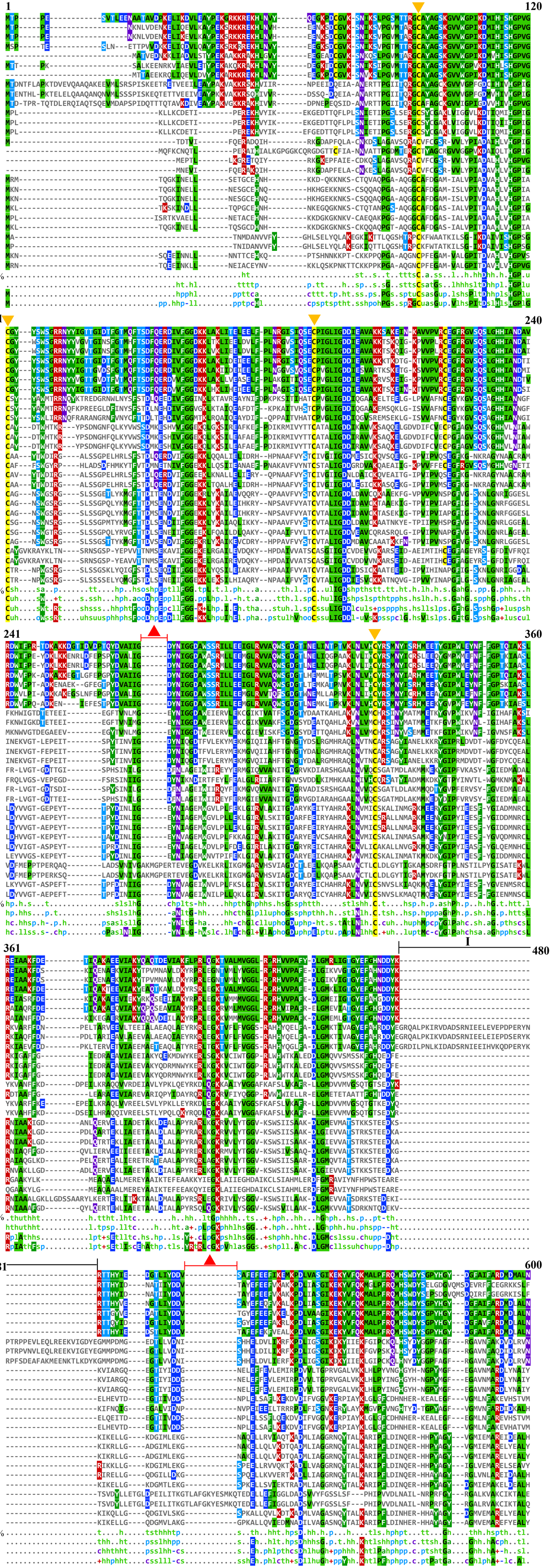	cov	identity
1 *Leptolyngbya boryana* dg5 NifD	100.0%	100.0%
2 *Nostoc* sp. PCC 7120 NifD	96.6%	73.0%
3 *Trichormus_variabilis* ATCC 29413 *nif*-cluster1 NifD	96.4%	74.0%
4 *Trichormus_variabilis* ATCC 29413_*nif*-cluster2 NifD	99.4%	85.8%
5 *Crocosphaera subtropica* ATCC 51142 NifD	97.2%	79.9%
6 *Synechococcus* sp. JA-3-3Ab NifD	98.6%	77.3%
7 *Hydrococcus rivularis* NIES-593 *nif*-cluster1 NifD	96.8%	83.0%
8 *Hydrococcus rivularis* NIES-593 *nif*-cluster2 NifD	98.4%	37.7%
9 Pleurocapsales cyanobacterium LEGE 06147 NifD	97.8%	37.7%
10 *Phormidium* sp. PBR-2020 NifD	98.2%	37.5%
11 *Trichormus_variabilis* ATCC 29413 VnfDG	92.3%	25.1%
12 *Nostoc carneum* NIES-2107 VnfDG1	92.3%	25.0%
13 *Nostoc carneum* NIES-2107 VnfDG2	92.5%	22.2%
14 *Hydrococcus rivularis* NIES-593 nif-cluster2 NifE	89.7%	27.8%
15 *Roseiflexus* sp. RS-1 NifD	90.9%	34.9%
16 *Phormidium* sp. PBR-2020 NifE	89.9%	28.6%
17 Pleurocapsales cyanobacterium LEGE 06147 NifE	89.5%	28.6%
18 *Leptolyngbya boryana* dg5 NifE	91.5%	29.2%
19 *Nostoc* sp. PCC 7120 NifE	91.7%	27.8%
20 *Trichormus variabilis* ATCC29413 nif-cluster1 NifE	91.7%	27.4%
21 *Crocosphaera subtropica* ATCC 51142 NifE	91.7%	27.3%
22 *Synechococcus* sp. JA-3-3Ab NifE	91.9%	26.8%
23 *Hydrococcus rivularis* NIES-593 nif-cluster1 NifE	91.7%	26.1%
24 *Trichormus variabilis* ATCC29413 VnfE1	88.9%	19.0%
25 *Nostoc carneum* NIES-2107 VnfE1	88.9%	18.6%
26 *Nostoc carneum* NIES-2107 NifE2/VnfE2	91.7%	26.3%
27 *Trichormus variabilis* ATCC29413 NifE4/VnfE2	91.5%	28.6%
consensus/100%		
consensus 90%		
consensus 80%		
consensus 70%